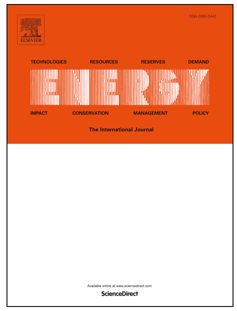
Accepted Manuscript



Pressure drop in the thermochemical recuperators filled with the catalysts of various shapes: A combined experimental and numerical investigation

Dmitry Pashchenko

PII: S0360-5442(18)32074-7

DOI: https://doi.org/10.1016/j.energy.2018.10.084

Reference: EGY 13981

To appear in: *Energy*

Received Date: 9 August 2018

Revised Date: 9 October 2018

Accepted Date: 15 October 2018

Pressure drop in the thermochemical recuperators filled with the catalysts of various shapes: a combined experimental and numerical investigation

Dmitry Pashchenko

Samara State Technical University
244 Molodogvardejskaya str., Samara 443100, Russia

Abstract

In this study, numerical simulations and experiments have been carried out to explore the the pressure drop and the loss factor in the thermochemical recuperators filled with the catalysts of various shapes. The reaction space of the thermochemical recuperators is presented as the porous packed bed with Ni-Al_2O_3 catalyst. The commercial program ANSYS Fluent is used. The comparison of the pressure drops between experimental and simulation results showed a good correlation with divergence of results less than 8%. The dependence of the pressure loss for the different depths of the packed bed is approximately linear. To determine the effect of the porosity properties of the medium on numerical results, two cases of CFD modeling were realized (with taking into account the porous medium properties and without it). The discrepancy between results is increasing with an increase of the gas velocity, while for the low velocities the results are almost similar. A recommendation for the engineering solutions is given, for saving calculation time at the low velocity, the packed bed can be considered as a solid body and the porous media properties can be ignored in the numerical model. Recommendations for engineering calculations of the thermochemical recuperation system are given.

Keywords: Thermochemical recuperation, Catalyst, Packed bed, CFD modeling, Pressure drop, Particle shape

Email address: pashchenkodmitry@mail.ru (Dmitry Pashchenko)

Nomenclature

ρ	density, kg/m^3;
v	velocity, m/s;
p	static pressure, Pa;
g	gravitational acceleration, m/s^2;
μ	molecular viscosity, Pa·s;
D_p	length of the fixed bed, m;
k	turbulence kinetic energy, J/kg;
ϵ	dissipation of turbulence energy, J/(kg·s);
$\alpha_\epsilon, \alpha_k$	inverse effective Prandtl numbers;
G_k	generation of turbulence kinetic energy due to the mean velocity gradients;
G_b	generation of turbulence kinetic energy due to buoyancy;
Y_M	contribution of the fluctuating dilatation in compressible turbulence;
S_k, S_ϵ	user-defined source terms;
$C_{1\epsilon}, C_{2\epsilon}$	enthalpy of hot flue gases after combustion chamber;
ϵ_v	void fraction of the packed bed;
CFD	computational fluid dynamics;
TCR	thermochemical recuperation.

1. Introduction

The problems of rational use of energy resources are becoming increasingly important because of the reduction in the world's fuel and energy reserves. These problems are particularly acute for plants with low energy efficiency. One of the main energy resources consumers are the industrial furnaces heated by natural gas [1, 2]. These furnaces designed for the glass production, steel heating, ceramics manufacture, etc, as a rule, generally show waste gas heat losses of about 25-30% of the total furnace energy input [1, 3, 4].

To improve energy efficiency, the furnaces are usually combined with heat-recovery systems. The conventional methods to increase the energy efficiency of the furnaces are the air and fuel preheating by the waste-heat gases before they enters the combustion chamber [5–7].

In recent decades, the recovery methods based on thermochemical transformation of the waste gases enthalpy to the chemical energy of a new synthetic fuel has found increasing interest among engineers and specialists [1, 8–14]. This method of waste-heat recovery is called thermochemical recuperation (TCR) [15]. The concept of increasing energy efficiency of fuel-consuming equipment by thermochemical waste-heat recuperation was discussed as early as 1960s [16]. A more detailed historical retrospective on the use of thermochemical recuperation is described in the comprehensive review presented by Tartakovsky and Sheintuch [17]. Generally, various chemical reforming reactions could be used for thermochemical waste-heat recuperation. Herewith, various endothermic and exothermic reactions (both desirable and undesirable) take place simultaneously. For example, for TCR coulbe be used steam methane reforming, which is described by the following chemical reactions:

$$CH_4 + H_2O \Leftrightarrow CO + 3H_2 + 206.1\ kJ; \tag{1}$$

$$CH_4 + 2H_2O \Leftrightarrow CO_2 + 4H_2 + 165.0\ kJ; \tag{2}$$

$$CO + H_2O \Leftrightarrow CO_2 + H_2 - 41.1\ kJ. \tag{3}$$

Moreover, there are publications that consider the use of methanol [18], ethanol [19, 20], propane [10], diesel [21], etc., for thermochemical recovery. The choice of the initial reaction depends on exhaust heat conditions (temperature, pressure, oxidizers). Therefore, selection of a most suitable primary fuel is a complex process [17].

Many thermochemical recovery investigations have focused on the industrial gas furnaces applications [2, 8, 22–24], engine applications [12, 25, 26], etc. Popov et al. showed that the energy efficiency of high-temperature plants with the thermochemical waste-heat recuperation was approximately 20 percentage points higher than that of the high-temperature plants without TCR [1].

Kobayashi [27] patented the thermochemical regenerative heat recovery process for industrial furnaces. A furnace is provided having at least two regenerator beds for heat recovery. While a first bed is being heated by hot flue gases produced by fuel combustion in the furnace, a second bed, heated during a previous cycle, is cooled through carrying out an endothermic chemical reforming reaction (for example, steam methane or dry methane reforming). Once the second bed is cooled by the endothermic reaction, the hot flue gases are redirected to it while the first bed, now hot, is used for carrying out the endothermic chemical reaction. Thereafter the cycle is repeated. Barati and co-authors [28] studied the method of utilization of thermal energy of the metallurgical slag by thermochemical recuperation. Also, thermochemical waste-heat recuperation via fuel reforming can be considered as on-board hydrogen production technology [12, 29–31].

Rue D. and co-workers [11] calculated the energy efficiency increasing of glass furnace with thermochemical recuperation. They reported that they have created the model of thermochemical waste-heat recuperation and tested it. The result of their work is the fact that energy efficiency of furnace with TCR is significantly higher than it is with thermal recuperators or other exchangers. Besides, an investigation of glass furnaces with TCR with steam methane reforming have been published in other articles [32–35]. In these papers the process conditions (temperature, pressure, inlet feed ratio) for syngas production have been studied. It was reported that complete conversion of methane can be achieved for temperatures above 700°C at inlet feed steam/methane ratio of about 3.0.

In addition, Chakravarthy et al. published the results of thermodynamic analysis of an internal combustion engine (ICE) with thermochemical exhaust heat recuperation [12]. The authors concluded that the estimated second law efficiency can be increased for TCR with ethanol and isooctane reforming by 9 and 11%, respectively. In addition, there are a series of contributions that also consider the use of TCR in the ICE [25, 26, 30].

Research studies on thermochemical recovery mentioned above were mainly focused on investigation of various schematic diagrams and thermodynamic analysis of original fuel reforming. The main attention is paid to heat and mass exchange processes in a thermochemical recuperator. The packed bed of reformers as well as monolith type can be used for TCR purposes [17]. However, it is known that the catalyst bed has a large aerodynamic resistance. The packed bed is typically used for an industrial-scale steam methane reformer [36]. More accurate knowledge of the flow dynamics in the thermochemical recuperators with the packed bed will make it possible to perform a more detailed analysis of the energy efficiency of the use of the thermochemical recuperation system in the industrial furnaces.

The aim of this study is to determine the aerodynamic resistance of the reaction space of the thermochemical recuperators filled with the catalysts of various shapes. For this aim, a computational fluid dynamics (CFD) model of flow dynamics in the reaction space of the recuperator taking into account porosity of the catalysts particles was developed.

With the modern computational tools and high power computing technology, the computational fluid dynamics simulations to predict the pressure drop of the reaction space of the thermochemical recuperators are feasible. Whether they will partly or fully replace experimental investigations will largely depend on the reliability of the turbulence and porous medium models that used, and the comprehensive study of it. Therefore, both numerical and experimental investigation of the pressure drop in the thermochemica is a crucial task.

2. Experiments

2.1. Experimental setup

To study fluid dynamics and the pressure drop of the thermochemical recuperator, the experimental setup shown in Fig. 1 is used. The experimental setup mainly consists of the following elements: 1 - a flow-meter and an electric heater ; 2 - the manometer tubes; 3 - a cylindrical tube with inner acrylic tube;

4 - an instrument board presenting the pressure and flow temperature; 5 - a transparent acrylic tube with catalysts; 6 - a fan/ventilator

To visualize the packing structure of a catalyst bed, the catalyst particles are placed to transparent acrylic tube. And then this tube is placed to the experimental setup. The gas flow is controlled by a fan rotor speed and measured by a Veterok GRM-12 flow-meter. The electronic manometers are used to measure pressure before and after the packed bed. The measuring positions of the electronic manometers were more than 3 times higher than the tube diameter to ensure stable pressure data.

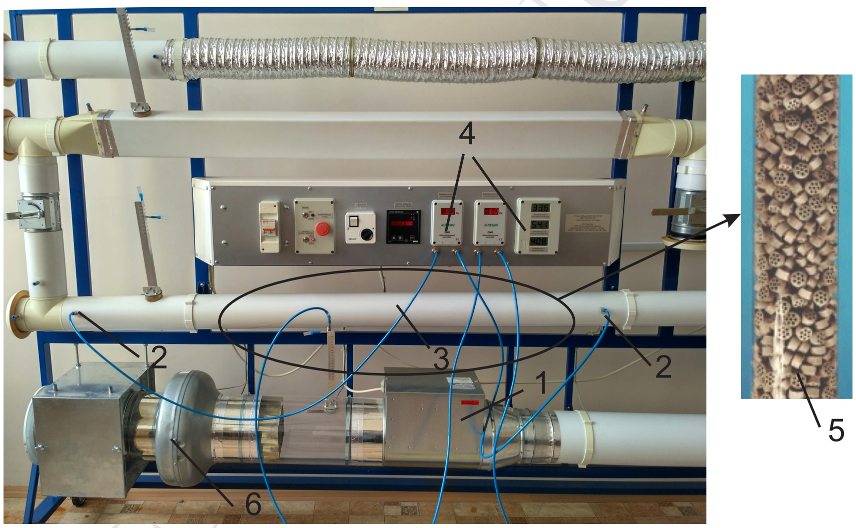


Figure 1: Photograph of a laboratory experimental setup for investigation of fluid dynamics and pressure drop

Each structure of the packed bed is tested under 10-15 different flow rates. For each flow rate, the measurements are repeated at least 3-5 times to quantify experimental precision. The pressure drop is determined as the average of the repetitions. For all experiments, air is heated by an electric heater up to 300°C. Air heating is necessary to bring the density of the air flow to the average density of the synthesis gas in the real steam methane reformer. All experiments are performed at ambient temperature (+20°C).

2.2. Catalyst shapes and packed bed

The packed beds with structured packing of catalyst particles with the different shapes are presented in Fig.2. There are four particle shapes: A – a simple cylinder; B – a Raschig ring (cylinder with one hole on axial line); C – a convex cylinder with 7 internal holes; D – a sphere with 7 internal holes.

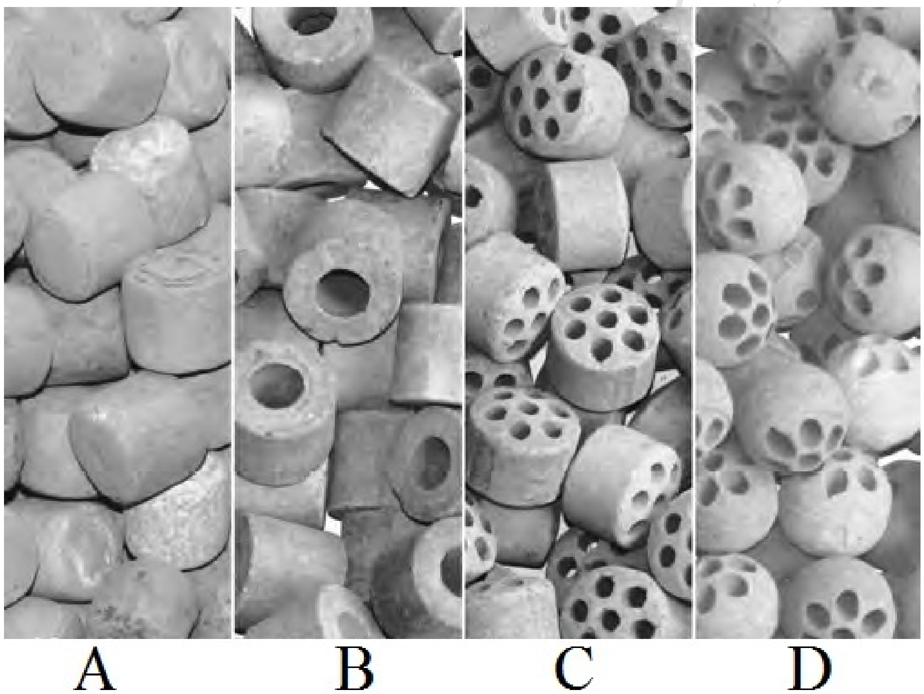


Figure 2: Photograph of the packed beds filled by the catalysts with different shapes: A – a cylindrical; B – a Raschig ring; C – a convex cylinder with 7 internal holes; D – a sphere with 7 internal holes

The catalyst particles are placed into the tube with diameter of 100 mm. The catalysts layer depth is 600 mm. The performances of catalysts are shown in Table 2. For experiment the Ni-Al_2O_3 catalyst is used with the following chemical composition: NiO = 14.5%; SiO_2 = 0.2%; support CaO-MgO-La_2O_3-Al_2O_3. Other properties are presented below: bulk density is 680 kg/m^3; surface area is 4.5 m^2/g$_{cat}$; average porosity of catalyst particles is about 41%. The last column in Table 1 shows the void fraction of the packed bed (external porosity).

The total porosity of the packed bed can be determined as the multiplication of the external and internal porosity.

Table 1: The geometrical characteristics of catalyst

Catalyst shape	Diameter, mm	Thickness, mm	Holes, number/diameter	Void fraction of packed bed
A	25	25	–	0.35
B	25	25	1x13	0.48
C	25	25	7x5	0.52
D	25	–	7x5	0.62

It is known that the effectiveness factor substantially depends on catalyst pellets size and catalyst coating technology [14]. In this study, only one type of the commercial catalyst (Ni-αAl_2O_3) and four particle shapes (Fig.2) is investigated. The catalyst is produced by NIAP-CATALYZATOR LLC (Novomoskovsk, Russia) [37].

In the experiments and numerical simulation, the same numbers of catalyst particles were used in such a way that the depth of the packed bed was the same both for the experiments and numerical simulation. Catalyst loading into the experimental setup is random. To generate the computational domain of the packed bed, RFM function in SolidWorks (Random falling or Random repositioning) was used. The packed bed for the experiment may differ from the packed bed for numerical simulation. However, these differences can be considered negligible, since the deep packed bed (600 mm) is used.

3. CFD modeling

3.1. Geometry and mesh

In the present paper, the family of particle shapes is considered as it is shown in Fig.2.

The computational domain has the maximum similarity to the experimental package of catalysts in the thermochemical recuperator. For this numerical

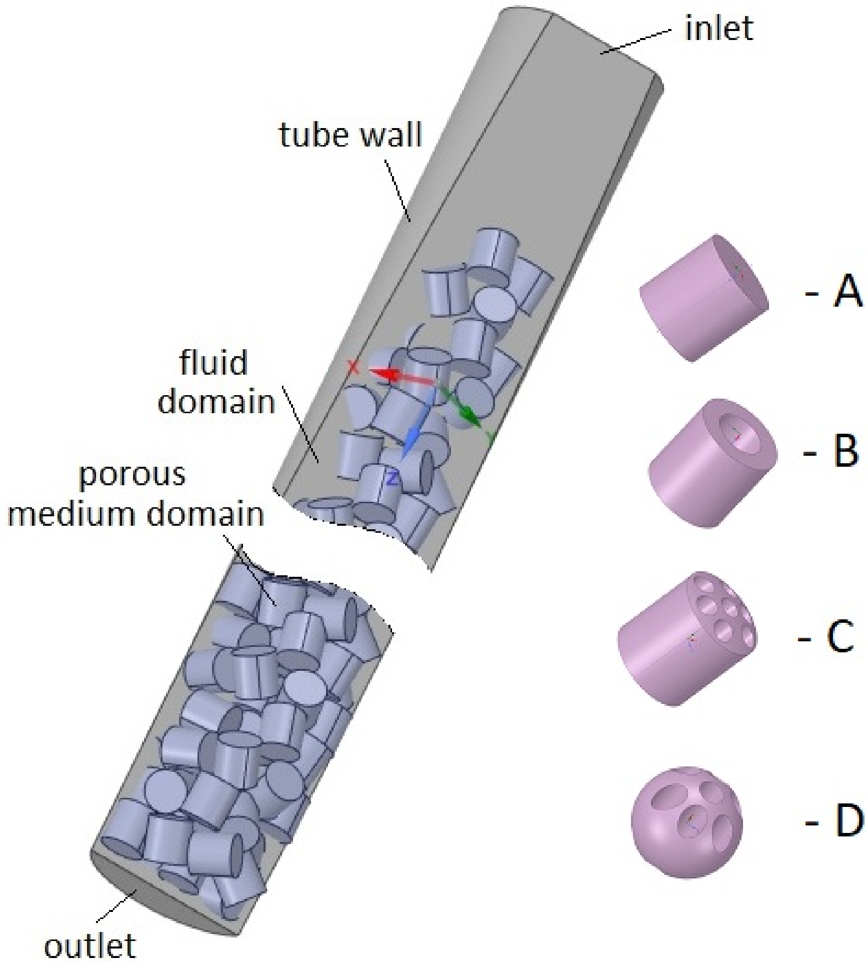


Figure 3: A computational domain of a packed bed that filled by the catalyst with cylindrical shape (A).

investigation, it is assumed that the size of all catalyst particles is the same. In reality, the diameter of the sphere can be different by 0.2-0.4 mm. The computational domain of the packed bed is presented in Fig.3. The packed bed is 600 mm in length, the diameter is 100 mm. All the cases have same outer diameter and length. To obtain a developed flow for experiments, when a laminar flow takes place, free space is added before the packed layer. The length of this free space is 250 mm.

The mesh structures for the all investigated computational domains presented in Fig.4. The edge size of the elementary cells is the same for the all computational domains. The meshes for the all cases are generated with the same mesh settings. To get a more perfect grid, the inflation layers around the catalyst were added in the fluid domain. Moreover, the inflation layer is also added near the tube wall in the fluid domain. The total thickness of the inflation layer is adjusted to get $20<y^+<250$, a requirement for k-ϵ turbulence model. The mesh independence study is not performed. The maximum number of elementary cells (about 23 millions for our university ANSYS license) are used.

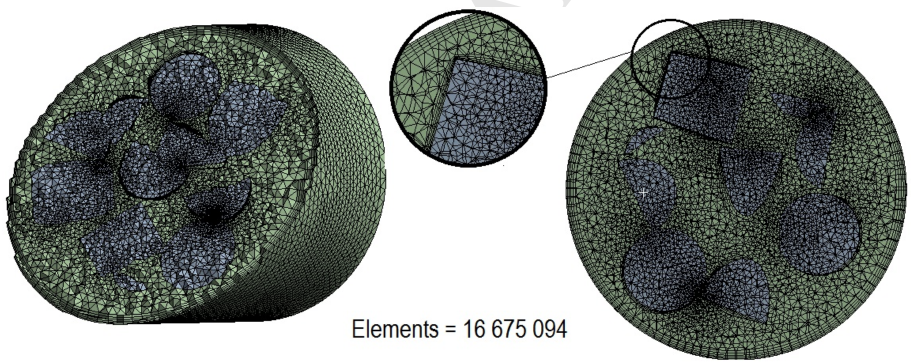


Figure 4: The mesh structure: a) catalyst porous media domain; b) fluid domain

3.2. Governing equations

Single phase, turbulent, steady flow are modeled by solving the Reynolds-averaged NavierStokes equations. The governing equations are solved in ANSYS Fluent. The assumptions are taken for CFD-modeling: the steady-state conditions; no flux of energy due to a mass concentration gradient (no Duflor effects) [38]; work by viscous forces and by pressure is not done.

The equation for the conservation of mass, or continuity equation, for steady-state conditions can be written as follows:

$$\nabla \cdot (\rho \vec{v}) = 0 \tag{4}$$

For the fluid domain the conservation of momentum is described by the following equations [39]:

$$\nabla \cdot (\rho \vec{v} \vec{v}) = -\nabla \cdot p + \nabla \cdot (\overline{\overline{\tau}}) + \rho \vec{g} + \vec{F} \tag{5}$$

where p is the static pressure; $\rho \vec{g}$ is the gravitational body force.

The stress tensor $\overline{\overline{\tau}}$ is given by the following equation:

$$\overline{\overline{\tau}} = \mu \left[(\nabla \vec{v} + \vec{v}^{\mathrm{T}}) - \frac{2}{3} \nabla \cdot \vec{v} I \right] \tag{6}$$

where μ is the molecular viscosity, I is the unit tensor, and the second term on the right hand side is the effect of volume dilation.

The porous media domain are simulated by the addition of source term for the equation (5) that can be expressed as:

$$S_i = -\left(\sum_{j=1}^{3} D_{ij} \mu v_j + \sum_{j=1}^{3} C_{ij} \frac{1}{2} \rho |v| v_j \right) \tag{7}$$

The momentum source term S_i for every dimension (x, y, z) consists of two parts. The first term of the equation (7) is a viscous loss term. The second term of the equation (7) is an inertial loss term. This momentum sink contributes to the pressure gradient in the porous cell, creating the pressure drop that is proportional to the fluid velocity (or velocity squared) in the cell.

The presented model is solving the standard conservation equations for turbulence quantities in the porous medium. In this default approach, turbulence in the medium is treated as though the solid medium has no effect on the turbulence generation or dissipation rates. This assumption may be reasonable if the medium's permeability is quite large and the geometric scale of the medium does not interact with the scale of the turbulent eddies.

Typically, steam methane reforming is taking place at a wide range of Reynolds numbers. RANS models offer the most economic approach for computing complex turbulent flows for a wide range of Reynolds numbers. A typical example of such model is k-ϵ model in different forms. The k-ϵ model is historically the most widely used turbulence models in industrial CFD-modeling. The

k-ϵ model in ANSYS Fluent falls within this class of models and has become the widely used of practical engineering flow calculations in the time since it was proposed by Launder and Spalding [40]. For the presented numerical model, the RNG k-ϵ model is chosen [41–44]. The Scalable Wall Function is used for RNG k-ϵ model, because there are large numbers of contacting surfaces in the computational domain. The mathematical description of RNG k-ϵ model can be expressed as following:

$$\frac{\partial}{\partial x_i}(\rho k v_i) = \frac{\partial}{\partial x_j}\left(\alpha_k \mu_{eff}\frac{\partial k}{\partial x_j}\right) + G_k + G_b - \rho\epsilon - Y_M + S_k \tag{8}$$

and

$$\frac{\partial}{\partial x_i}(\rho \epsilon v_i) = \frac{\partial}{\partial x_j}\left(\alpha_\epsilon \mu_{eff}\frac{\partial \epsilon}{\partial x_j}\right) + C_{1\epsilon}\frac{\epsilon}{k}(G_k + C_{3\epsilon}G_b) - C_{2\epsilon}\rho\frac{\epsilon^2}{k} - R_\epsilon + S_\epsilon \tag{9}$$

In equations for k-ϵ turbulence model, G_k shows the generation of turbulence kinetic energy due to the mean velocity gradients. G_b is the generation of turbulence kinetic energy due to buoyancy. Y_M represents the contribution of the fluctuating dilatation in compressible turbulence to the overall dissipation rate. The quantities α_k and α_ϵ are the inverse effective Prandtl numbers for k and ϵ, respectively. S_k and S_ϵ are user-defined source terms.

The additional term R_ϵ in the ϵ equation can be determined as follows:

$$R_\epsilon = \frac{C_\mu \rho \eta^3 (1 - \eta/\eta_0)}{1 + \beta\eta^3}\frac{\epsilon^2}{k} \tag{10}$$

where $\eta \equiv Sk/\epsilon$, $\eta_0 = 4.38$, $\beta = 0.012$.

The model constants $C_{1\epsilon}$ and $C_{2\epsilon}$ in k-ϵ turbulence model used by default setup for Fluent solver of ANSYS [45]:

$$C_{1\epsilon} = 1.42; \;\; C_{2\epsilon} = 1.68 \tag{11}$$

The performances of the porous medium are determined by the viscous resistance coefficients and the inertial resistance coefficients for each direction. These coefficients, respectively, can be determined as following:

$$\alpha = \frac{D_p^2}{150}\frac{\epsilon_v^3}{(1-\epsilon_v)^2}; \tag{12}$$

$$C_2 = \frac{1.75\rho}{D_p}\frac{(1-\epsilon_v)}{\epsilon_v^3}, \tag{13}$$

where D_p – the length of the fixed bed; ϵ – the void fraction. Here the void fraction (ϵ) is taken into account the interparticle spaces and the pellet porosity.

4. Results and discussion

4.1. Verification

To verify the numerical results, the experimental determination of the pressure drop in the thermochemical recuperator is performed for the various shapes of catalyst particles. In this study four shapes of the catalyst particles are considered: a cylinder, a cylinder with internal holes, a Raschig ring, a sphere with internal holes. The experimental and numerical study is performed for the different initial conditions. The commercial ANSYS Fluent CFD code is used for numerical simulation.

The experimental and numerical investigation were performed at same operating conditions. The CFD model was realized via ANSYS Fluent on Xeon E5-2670 Sandy Bridge-EP server with double 20-core 2.6 GHz processors with 32 GB RAM. Convergence for each numerical experiment took approximately 3000 iterations for which the computation time was on the order of 20-23 hours. The Reynolds number during both experimental and numerical investigations ranged from about 500 to about 9000. The change in the Reynolds number is done by changing the velocity.

Further, all the experimental data are denoted by the hollow points. The results of numerical simulation are represented by the solid lines. To verify the results obtained with the help of CFD model, they were compared with experimental data for the different packed beds. Fig.5 shows the comparison of numerical and experimental data for the different packed beds with 600 mm in length. The value of the pressure drop for the numerical results was obtained by Vortex Average function in Fluent solver.

As can be seen from Fig.5, the differences between the experimental and numerical results are observed for all investigated cases. However, at the velocity

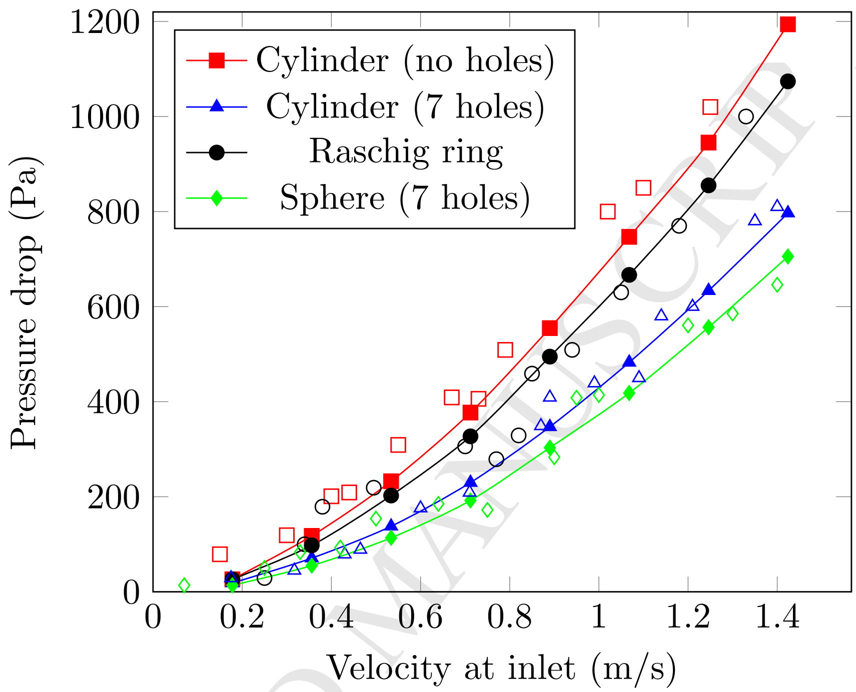


Figure 5: The comparison of numerical and experimental data: the hollow points – experiment; the solid points and lines – CFD modeling.

range above 0.5 m/s, which corresponds to the real operation parameters of the thermochemical recuperators, the discrepancies between the results are reduced. The average deviations between the numerical and experimental results in the velocity range above 1 m/s are 6.12% for the cylinder without holes, 2.15% for the Raschig ring, 4.75% for the cylinder with 7 holes, 4.34% for the sphere with 7 holes. Therefore, it can be concluded that the results have a good correlation and the mathematical model is reliable to study the flow dynamics in the packed bed.

4.2. Pressure and velocity contours

Fig.6 shows the typical CFD simulation results for the packed bed filled with the cylindrical catalysts, including the velocity contours on the catalyst particles (Fig.6A), the velocity contours into the catalyst particles (Fig.6B) and the pressure contours on particles (Fig.6C). Each point on the pressure profile plot corresponds to one spatial position in the packing. Fig.6 depicts that the flow of gas passes not only between the catalyst particles, but inside them.

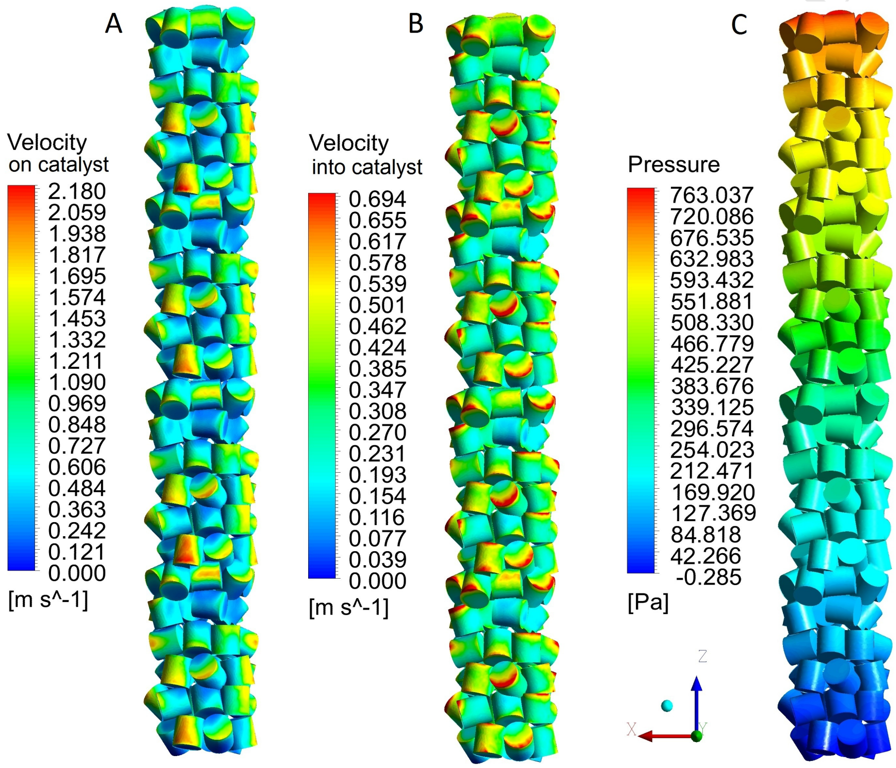


Figure 6: The velocity and pressure contours.

Fig.6 shows that the stream flow takes place not only between the catalyst particles, but inside the catalyst particles. This is due to the fact that the properties of the catalyst particles are set as a porous medium. In addition, from Fig.6, it can be seen that the velocity on the surface of the catalyst particles is much higher than the velocity inside the catalyst, because the viscous resistance coefficients and the inertial resistance coefficients are set in the CFD model. Analysis of the pressure contour (Fig.6(C)) shows that the pressure varies monotonically along the packed bed, thereby showing that the aerodynamic drag is directly proportional to the depth of the packed bed. This fact is discussed in this article below. Moreover, a uniform drop in pressure indicates that the random loading of catalyst particles is correct because there are no noticeable pressure surges.

The velocity contours in the axial and radial cross-section of the packed bed and are shown in Fig.7. The velocity magnitude is significantly lower into the particle shapes compared to the velocity magnitude into free space between the cylindrical particles.

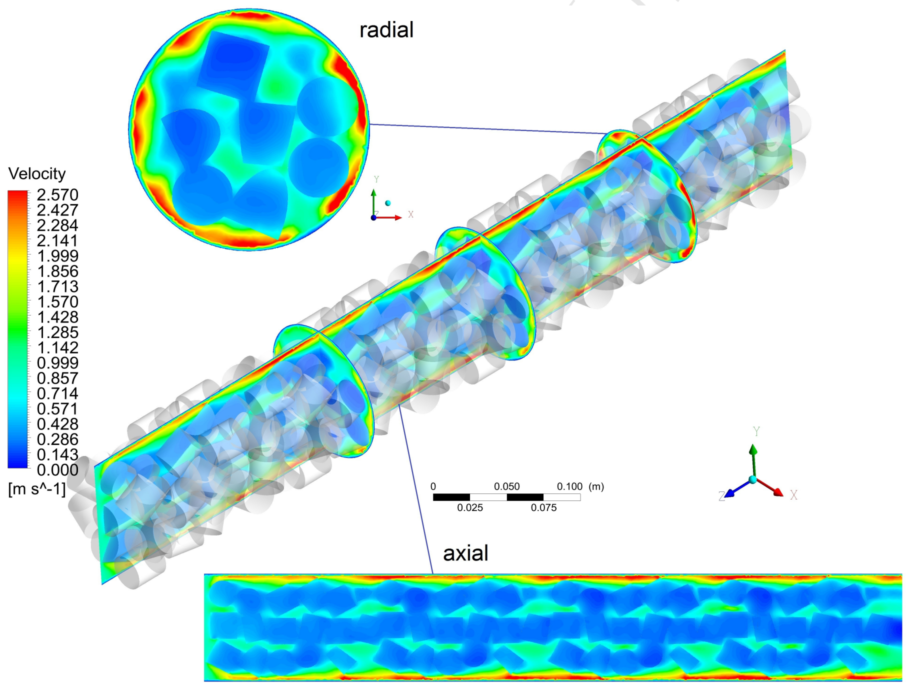


Figure 7: Simulated velocity magnitude distribution for velocity u=0.7 m/s.

4.3. Loss factor

Fig.8 shows the dependence of the pressure losses per meter of the packed bed for different Reynolds numbers. From Fig.8 it can be seen that the maximum pressure loss per 1 meter of the packed bed takes place for a cylinder without holes (case A in Table1), while a minimum pressure loss occurs for a cylinder with 7 holes (case C in Table1). At the same time, a characteristic quadratic dependence $\Delta P = f(Re^2)$ is observed for all four curves: $\Delta P = f(Re^{1.92})$ for A case; $\Delta P = f(Re^{1.94})$ for B case; $\Delta P = f(Re^{1.93})$ for C case; $\Delta P = f(Re^{1.94})$ for D case. Here, the Reynolds number is calculated for the inlet section of the tube (empty tube) with the packed beds.

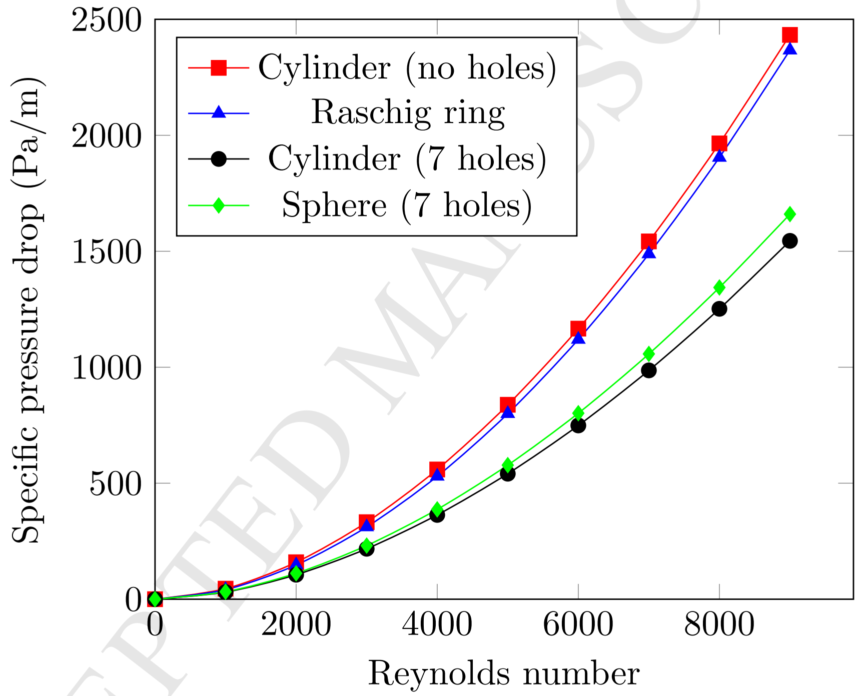


Figure 8: The variations of the pressure drop versus Reynolds number for the various packed beds.

From the point of view of the classical representation of fluid dynamics, the pressure losses in the catalyst bed can be considered as the local hydraulic losses, which are caused by changes in the shape and size of the channel, which is deforming the flow. As shown by the experimental and numerical results (Fig.5 and Fig.8), it can be approximately assumed that the pressure loss through the catalyst bed is proportional to the square of the velocity. For this reason, for the engineering calculations, it is convenient to characterize the fluid resistance of a catalyst bed with a dimensionless quantity K_l, which is called the local resistance coefficient or loss factor.

The local resistance coefficient can be determined experimentally:

$$K_l = \frac{2\Delta P}{\rho v^2}, \tag{14}$$

where ρ – the density of fluid; v – the velocity of fluid.

To determine the local resistance coefficient, the effect of length of the packed bed on the pressure drop is investigated. For this purpose, the different lengths of the packed bed are considered. The computational domains (Fig.3) were changed, but the mesh structures were same as in previous cases. Fig.9 shows the variations of the pressure drop as a function of the packed beds length for Re=6000.

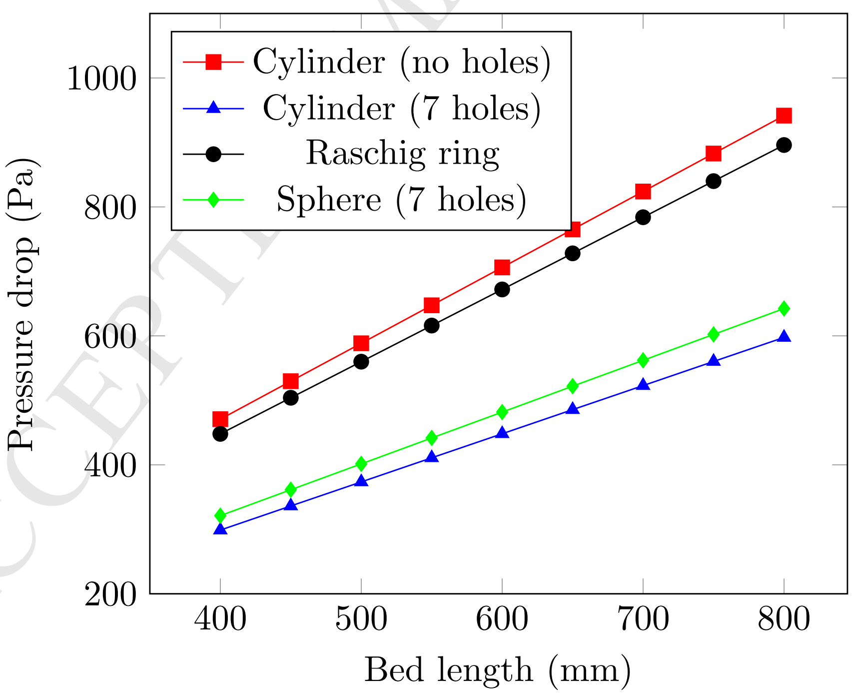


Figure 9: The variations of the pressure drop for the different packed beds length for Re=6000.

As it can be seen from Fig.9, the dependence of the pressure loss for the different lengths of the catalyst bed is approximately linear. This fact allows to conclude that the value of the local resistance coefficient for the catalyst bed is determined by the amount of catalyst in the packed bed and the shape of catalyst particle.

Based on the results which are presented in Fig. 9, and also on the basis of the expression (14), the values of the local resistance coefficients were determined for various cases (Table1). The variations of the local resistance coefficient for 1 meter of the packed bed versus the average velocity for the different packed beds are shown in Fig.10.

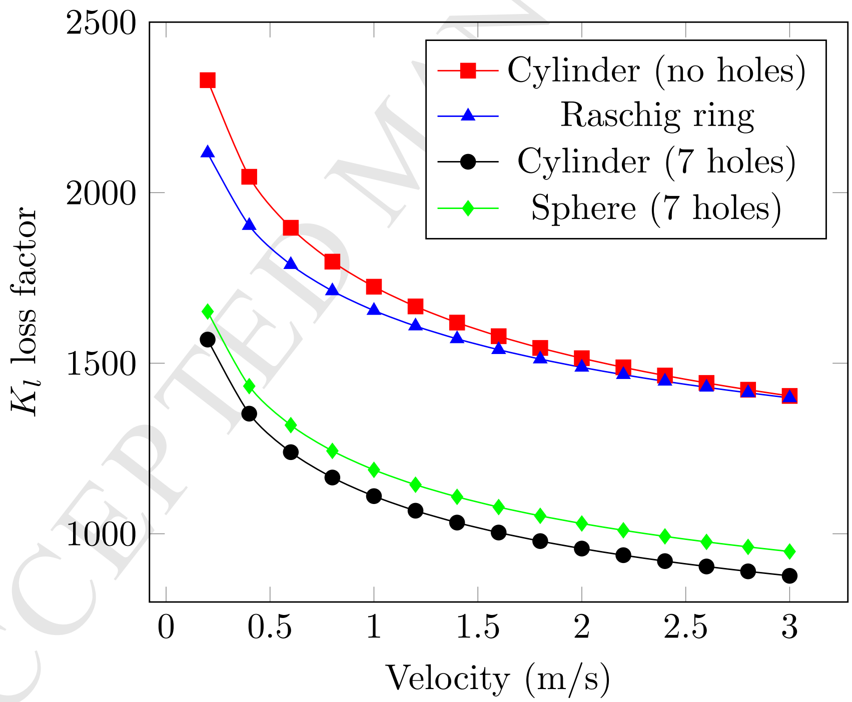


Figure 10: The variations of the local resistance coefficients versus the velocity for the different packed beds for Re=6000.

As seen in Fig.10, the loss factor K_l varies substantially when the velocity varies, especially in the range up to 1 m/s. Usually, when flowing around solid non-porous bodies, the coefficient K_l depends weakly on the velocity. This effect of velocity on the coefficient K_l can be explained by the fact that with increasing velocity, the amount of gas that passes through the pores in catalyst increases. At the same time, when the gas velocity is low, the fluid energy is not enough to percolate through the porous structure. The next paragraph presents the results of investigation the velocity distribution in porous media.

The value of the coefficient K_l in the general terms depends on the geometry (the form of local resistance, the relative roughness of the walls, the distribution of velocities in the boundary sections of the flow before and after the local resistance) and velocity. When velocity changes, the coefficient K_l value will remain constant (for the wide range of velocity). Therefore, it is possible to use the obtained values of K_l for calculating the pressure drop in the real recuperators when the density of the reaction mixture varies along the packed bed of the recuperator.

4.4. Porous media

There are scientific articles dealing with the study of flow dynamics in a packed bed that treat catalyst particles as a solid body [46]. In this paragraph, the effect of taking into account of the porous medium properties on the numerical results is reported. For this purpose, the computational domains that presented above were considered. The two type of the packed beds are considered: A – cylinder without holes; D – sphere with 7 holes.

Fig.11 show the discrepancy between the results obtained with taking into account of the porous medium properties (solid line) and without it (dashed line). As it can be seen from Fig.11, the discrepancy between the results is increasing with an increase of the velocity magnitude, while for the low velocity the results are almost similar. Therefore, for saving calculation time at low velocity, the packed bed can be considered as a solid body and the porous media properties can be ignored in the numerical model. However, for the velocity above 1 m/s, the porous medium properties must be taken into account, because the discrepancies are more than 10%.

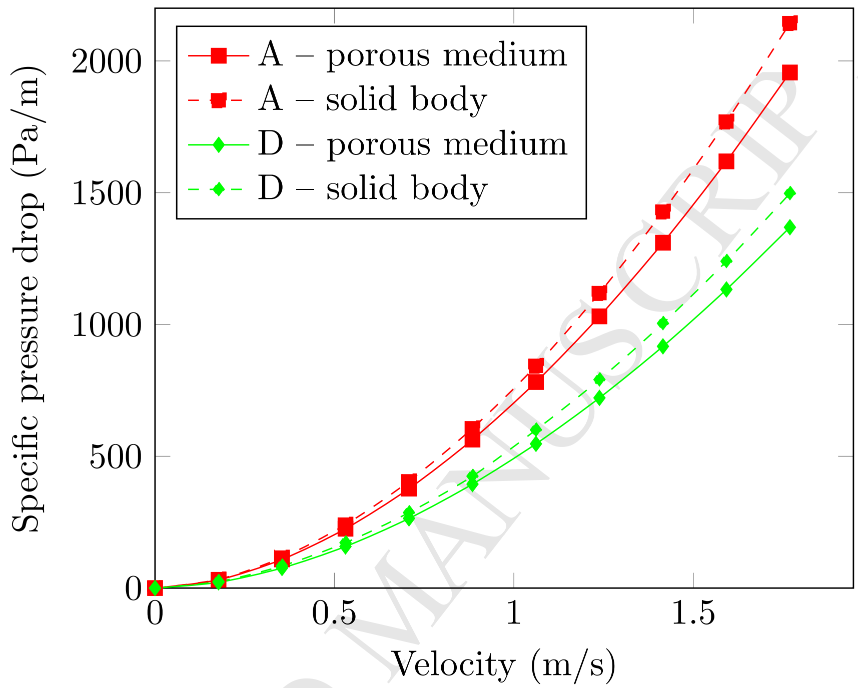


Figure 11: The variations of the specific pressure drop versus the velocity with taking into account of the porous medium properties and without it.

5. Conclusion

The results of numerical simulation and experimental investigation of fluid dynamics in the the thermochemical recuperators filled with the catalysts of various shapes are reported in this paper. The good correlation between the numerical and experimental results is achieved for all experiments with an average error of less than 8%.

A characteristic quadratic dependence $\Delta P = f(v^2)$ is observed for all catalyst types. The effect of the packed bed depth on the pressure drop is investigated. The dependence of the pressure loss for the different depths of the catalyst bed is approximately linear. This fact allowed to conclude that the local resistance coefficient for the catalyst bed is constant for all investigated L/d ratios. The loss factor K_l varies substantially when the velocity varies, especially in the range up to 1 m/s.

To determine the effect of porosity properties of the medium on numerical results, two cases of CFD modeling were realized. First, with taking into account the porous medium properties; second, without it. The discrepancy between the results is increasing with an increase of the velocity, while for the low velocity the results are almost similar. As a recommendation for the engineering solutions, it can be concluded that for saving calculation time at the low velocity, the packed bed of the the thermochemical recuperators can be considered as a solid body and the porous media properties can be ignored in the numerical model. However, for the velocity above 1 m/s, the porous medium properties must be taken into account, because the discrepancies are more than 10%.

References

[1] S. Popov, I. Svistunov, A. Garyaev, E. Serikov, E. Temyrkanova, The use of thermochemical recuperation in an industrial plant, Energy 127 (2017) 44–51.

[2] D. Pashchenko, First law energy analysis of thermochemical waste-heat recuperation by steam methane reforming, Energy 143 (2018) 478–487.

[3] R. G. Beerkens, H. P. Muysenberg, Comparative study on energy-saving technologies for glass furnaces., GLASTECH BER. 65 (8) (1992) 216–224.

[4] E. Worrell, L. Bernstein, J. Roy, L. Price, J. Harnisch, Industrial energy efficiency and climate change mitigation, Energy efficiency 2 (2) (2009) 109.

[5] Y. A. Cengel, R. H. Turner, J. M. Cimbala, M. Kanoglu, Fundamentals of thermal-fluid sciences, McGraw-Hill New York, 2008.

[6] M. A. A. Alfellag, Parametric investigation of a modified gas turbine power plant, Thermal Science and Engineering Progress 3 (2017) 141–149.

[7] K. Werner, Fuel preheating in a gas turbine, uS Patent 6,381,945 (May 7 2002).

[8] G. Verkhivker, V. Kravchenko, The use of chemical recuperation of heat in a power plant, Energy 29 (3) (2004) 379–388.

[9] C. Gaber, M. Demuth, R. Prieler, C. Schluckner, C. Hochenauer, An experimental study of a thermochemical regeneration waste heat recovery process using a reformer unit, Energy 155 (2018) 381–391.

[10] D. Pashchenko, Thermodynamic equilibrium analysis of combined dry and steam reforming of propane for thermochemical waste-heat recuperation, International Journal of Hydrogen Energy 42 (22) (2017) 14926–14935.

[11] D. Rue, A. Kozlov, M. Khinkis, H. Kurek, Thermochemical recuperation to increase glass furnace energy efficiency, in: 74th Conference on Glass Problems, Wiley Online Library, 2014, pp. 81–92.

[12] V. K. Chakravarthy, C. S. Daw, J. A. Pihl, J. C. Conklin, Study of the theoretical potential of thermochemical exhaust heat recuperation for internal combustion engines, Energy & Fuels 24 (3) (2010) 1529–1537.

[13] D. Pashchenko, Energy optimization analysis of a thermochemical exhaust gas recuperation system of a gas turbine unit, Energy Conversion and Management 171 (2018) 917–924.

[14] A. Poran, A. Thawko, A. Eyal, L. Tartakovsky, Direct injection internal combustion engine with high-pressure thermochemical recuperation–experimental study of the first prototype, International Journal of Hydrogen Energy 43 (27) (2018) 11969–11980.

[15] G. P. Olmsted, JH, Heat engine efficiency enhancement-through chemical recovery of waste heat, 7th Intersociety Energy Conversion Engg. Con (1972) 241–248.

[16] R. Lee, D. B. Wimmer, Exhaust emission abatement by fuel variations to produce lean combustion, SAE Transactions (1968) 3025–3044.

[17] L. Tartakovsky, M. Sheintuch, Fuel reforming in internal combustion engines, Progress in Energy and Combustion Science 67 (2018) 88–114.

[18] L. Tartakovsky, R. Amiel, V. Baibikov, R. Fleischman, M. Gutman, A. Poran, M. Veinblat, Si engine with direct injection of methanol reforming products-first experimental results, Tech. rep., SAE Technical Paper (2015).

[19] L. Tartakovsky, V. Baibikov, M. Gutman, A. Mosyak, M. Veinblat, Performance analysis of si engine fueled by ethanol steam reforming products, Tech. rep., SAE Technical Paper (2011).

[20] D. Pashchenko, Thermochemical recovery of heat contained in flue gases by means of bioethanol conversion, Thermal Engineering 60 (6) (2013) 438–443.

[21] F. D. Chuahy, S. L. Kokjohn, High efficiency dual-fuel combustion through thermochemical recovery and diesel reforming, Applied Energy 195 (2017) 503–522.

[22] R. Carapellucci, A. Milazzo, Thermodynamic optimization of a reheat chemically recuperated gas turbine, Energy Conversion and Management 46 (18-19) (2005) 2936–2953.

[23] D. Cocco, V. Tola, G. Cau, Performance evaluation of chemically recuperated gas turbine (crgt) power plants fuelled by di-methyl-ether (dme), Energy 31 (10-11) (2006) 1446–1458.

[24] R. Carapellucci, A unified approach to assess performance of different techniques for recovering exhaust heat from gas turbines, Energy Conversion and Management 50 (5) (2009) 1218–1226.

[25] L. Tartakovsky, A. Mosyak, Y. Zvirin, Energy analysis of ethanol steam reforming for hybrid electric vehicle, International Journal of Energy Research 37 (3) (2013) 259–267.

[26] A. Poran, L. Tartakovsky, Energy efficiency of a direct-injection internal combustion engine with high-pressure methanol steam reforming, Energy 88 (2015) 506–514.

[27] H. Kobayashi, Thermochemical regenerative heat recovery process, uS Patent 6,113,874 (Sep. 5 2000).

[28] M. Barati, S. Esfahani, T. Utigard, Energy recovery from high temperature slags, Energy 36 (9) (2011) 5440–5449.

[29] M. Bogarra, J. Herreros, A. Tsolakis, A. York, P. Millington, Study of particulate matter and gaseous emissions in gasoline direct injection engine using on-board exhaust gas fuel reforming, Applied Energy 180 (2016) 245–255.

[30] A. Poran, L. Tartakovsky, Performance and emissions of a direct injection internal combustion engine devised for joint operation with a high-pressure thermochemical recuperation system, Energy 124 (2017) 214–226.

[31] L. Zhu, Z. He, Z. Xu, X. Lu, J. Fang, W. Zhang, Z. Huang, In-cylinder thermochemical fuel reforming (tfr) in a spark-ignition natural gas engine, Proceedings of the Combustion Institute 36 (3) (2017) 3487–3497.

[32] H. van Limpt, R. Beerkens, A. Habraken, Energy recovery from waste heat in the glass industry & thermo chemical recuperator, in: 73rd Conference on Glass Problems: Ceramic Engineering and Science Proceedings, John Wiley & Sons, New York, USA, Vol. 34, 2013, p. 3.

[33] F. J. Nelson, Waste heat recovery from regenerative glass furnaces using an air extraction process, in: 51st Conference on Glass Problems: Ceramic Engineering and Science Proceedings, Volume 12, no. 3-4, John Wiley & Sons, 2009, p. 568.

[34] Y. Y. Pechenegov, V. Simonov, B. Semenov, O. Y. Kosova, A. Mrakin, Oxidative pyrolysis of oil shale in tubular flow reactors with external heating, Solid Fuel Chemistry 51 (1) (2017) 40–43.

[35] S. Sikirica, H. Kurek, A. Kozlov, M. Khinkis, Thermochemical recuperation improves furnace thermal efficiency, Heat Treating Progress 7 (5) (2007) 28.

[36] A. Tran, A. Aguirre, H. Durand, M. Crose, P. D. Christofides, CFD modeling of a industrial-scale steam methane reforming furnace, Chemical Engineering Science 171 (2017) 576–598.

[37] D. Pashchenko, Numerical study of steam methane reforming over a preheated ni-based catalyst with detailed fluid dynamics, Fuel 236 (2019) 686–694.

[38] M. Lücke, et al., Influence of the Dufour effect on convection in binary gas mixtures, Physical review E 52 (1) (1995) 642.

[39] G. K. Batchelor, An introduction to fluid dynamics, Cambridge university press, 2000.

[40] B. E. Launder, D. B. Spalding, The numerical computation of turbulent flows, in: Numerical Prediction of Flow, Heat Transfer, Turbulence and Combustion, Elsevier, 1983, pp. 96–116.

[41] H. Yilmaz, O. Cam, S. Tangoz, I. Yilmaz, Effect of different turbulence models on combustion and emission characteristics of hydrogen/air flames, International Journal of Hydrogen Energy.

[42] D. Pashchenko, ANSYS fluent CFD modeling of solar air-heater thermoaerodynamics, Applied Solar Energy 54 (1) (2018) 32–39.

[43] D. Pashchenko, Comparative analysis of hydrogen/air combustion cfd-modeling for 3d and 2d computational domain of micro-cylindrical combustor, International Journal of Hydrogen Energy 42 (49) (2017) 29545–29556.

[44] A. Mrakin, A. Selivanov, P. Batrakov, D. Sotnikov, The chemical energy unit partial oxidation reactor operation simulation modeling, in: Journal of Physics: Conference Series, Vol. 944, IOP Publishing, 2018, p. 012082.

[45] V. Yakhot, S. A. Orszag, Renormalization group analysis of turbulence. I. Basic theory, Journal of scientific computing 1 (1) (1986) 3–51.

[46] H. Bai, J. Theuerkauf, P. A. Gillis, P. M. Witt, A coupled DEM and CFD simulation of flow field and pressure drop in fixed bed reactor with randomly packed catalyst particles, Industrial & Engineering Chemistry Research 48 (8) (2009) 4060–4074.

Experimental data and numerical results obtained with the help of high-performance computing (HPC)

The pressure losses in the catalyst bed can be considered as the local hydraulic losses.

The loss factor K_l varies substantially when the velocity varies.

The packed bed can be considered as a solid body for the low velocity below 1m/s.

The porous medium properties must be taken into account for velocity above 1m/s.